Unified Multi-Domain Process Network Analysis Platform (UMPNAP)

Detailed Design and Concept Proposal

Executive Summary

This document proposes the development of an indigenous Unified Multi-Domain Process Network Analysis Platform (UMPNAP). The platform is intended to provide independent engineering analysis capability for process systems and to complement existing simulator development activities. The platform shall operate completely offline on Red Hat Enterprise Linux or Bharat Linux systems and shall provide a Qt-based graphical environment for network construction, configuration, simulation, analysis, validation and reporting.

Problem Statement

Simulator teams frequently rely on external engineering analysis groups for thermo-hydraulic studies, pressure calculations, flow distribution studies, heat transfer assessments and accident-condition evaluations. This dependency can slow simulator validation, limit engineering experimentation and reduce internal ownership of process modelling knowledge. A dedicated analysis platform can provide an independent reference for simulator validation and engineering studies.

Primary Objectives

1. Provide a graphical P&ID-based; modelling environment. 2. Support hydraulic, gas, thermal, steam and electrical domains. 3. Allow mixed-domain modelling in a common workspace. 4. Support offline execution. 5. Provide independent validation capability for simulator models. 6. Create a reusable engineering platform for future simulator development. 7. Establish a foundation for future advanced thermal-hydraulic capabilities.

Target Use Cases

Moderator systems, cover gas systems, compressed air systems, service water systems, oil systems, cooling water systems, electrical auxiliary systems, pump studies, valve sizing studies, heat exchanger studies, transient response analysis, simulator validation, engineering sensitivity studies and training of new modelling engineers.

Key Functional Requirement

A user shall drag and drop components from a library, connect them graphically, configure parameters and execute analyses. The same workspace shall support D2O loops, helium cover gas spaces, air systems and thermal equipment. Separate workspaces for different fluids shall not be required.

System Architecture

The architecture shall consist of: 1. Qt-based graphical editor. 2. Project repository. 3. Model manager. 4. Solver manager. 5. Domain solvers. 6. Property package manager. 7. Results database. 8. Plotting and reporting framework. The GUI shall never directly solve equations. It shall only define the network topology and configuration.

Domain Solver Framework

A Solver Manager shall coordinate multiple solver engines. Hydraulic Solver: incompressible and mildly compressible liquids. Gas Solver: air, helium, nitrogen and other gases. Thermal Solver: heat transfer and thermal masses. Electrical Solver: electrical distribution studies. Coupled Solver: interaction between liquid and gas volumes. The solver manager shall assemble the global system of equations and coordinate iterative convergence.

Numerical Methods

The numerical engine shall support: - Newton-Raphson nonlinear solution. - Sparse matrix assembly. - LU factorization. - Iterative solvers. - Transient integration. - Mass conservation. - Momentum conservation. - Energy conservation. Future releases may include adaptive timestep control and parallel execution.

Generic Fluid Property Framework

Components shall not be hardcoded for a specific fluid. A central property package shall provide density, viscosity, specific heat, thermal conductivity, compressibility and other properties. The user shall be able to switch between Light Water, Heavy Water (D2O), Oil, Air, Helium and Nitrogen directly from the GUI. The same Pipe, Pump or Valve model shall automatically adapt to the selected fluid properties.

Component Library Architecture

Components shall be organised by domain: /components/hydraulic Pipe, Pump, Valve, Tank, Boundary, Junction, Orifice, Heat Exchanger /components/gas Gas Volume, Regulator, Compressor, Relief Valve /components/electrical Bus, Generator, Breaker, Transformer, Load /components/control PID, Logic, Timer, Comparator, Trips /components/thermal Thermal Mass, Heat Source, Heat Sink All components shall be plugin-based.

Graphical User Interface

The Qt interface shall support: - Drag-and-drop modelling. - Property editors. - Context menus. - Connection validation. - Model hierarchy browser. - Trend displays. - Alarm and error display. - Engineering unit selection. - Save and restore of complete projects.

Results and Visualisation

Users shall be able to trend: - Pressure - Flow - Temperature - Tank Level - Inventory - Valve Position - Pump Head - Heat Duty The platform shall provide real-time plots during execution and post-processing analysis after execution.

Moderator System Example

A complete moderator system model shall include: - Calandria vessel. - Heavy water inventory. - Moderator pumps. - Heat exchangers. - Connecting pipework. - Helium cover gas region. The gas and liquid portions shall exist in the same workspace and be solved as a coupled system.

Validation Strategy

Validation shall occur at multiple levels: 1. Unit testing of individual components. 2. Verification against analytical solutions. 3. Comparison against plant design calculations. 4. Comparison against simulator behaviour. 5. Benchmarking against accepted engineering tools where possible.

Development Roadmap

Phase 1: Single-phase hydraulic solver. Phase 2: Thermal framework and heat exchangers. Phase 3: Gas network solver. Phase 4: Coupled gas-liquid systems. Phase 5: Steam and two-phase thermal hydraulics. Phase 6: Advanced engineering studies and optimization.

Expected Benefits

Reduced dependence on external analysis tools, improved simulator validation capability, development of indigenous engineering expertise, reusable software assets, knowledge retention, improved development efficiency and a foundation for future advanced process modelling platforms.